

OPTIKA OS

Optika biznesini boshqarish tizimi

Professional Technical Specification / TZ v2.0

1. LOYIHA KONSEPSIYASI

Optika OS — optika do'koni uchun:

- CRM;
- POS;
- ombor;
- buyurtmalar;
- retseptlar;
- kassa;
- supplier;
- qarzdorlik;
- hisobot;
- xodimlar;
- audit;
- avtomatik bildirishnomalar

tizimlarini bitta platformada birlashtiradigan dastur.

Tizimning asosiy maqsadi:

Xodimga imkon qadar kam tugma bosdirib, optikaning barcha operatsiyalarini avtomatik hisobga olish.

2. ASOSIY PRINSIP

Tizim murakkab ERP ko'rinishida bo'lmasligi kerak.

Xodim quyidagilar haqida o'ylamasligi kerak:

- qaysi omborni tanlash;
- qaysi kassani tanlash;
- qaysi filialni tanlash;
- mahsulot kodini eslab qolish;
- mijoz ID sini bilish;

- qoldiqni qo'lda ayirish;
- qarzni qo'lda hisoblash;
- buyurtma statusini qo'lda yuritish.

Tizim bularni avtomatik bajaradi.

3. PLATFORMALAR

Asosiy

Responsive Web Application / PWA.

Ishlaydi:

- Desktop;
- Laptop;
- Tablet;
- Smartphone.

Desktop versiya asosiy hisoblanadi.

Telefon versiyasi:

- tezkor qidiruv;
- dashboard;
- buyurtma holati;
- ombor;
- rahbar nazorati

uchun optimallashtiriladi.

4. TEXNOLOGIK STACK

Frontend

Next.js / React

TypeScript

TanStack Query

React Hook Form

Zod

PWA support

Backend

FastAPI

Python

SQLAlchemy

Alembic

Pydantic

Database

PostgreSQL

Cache

Redis

Background Jobs

Celery / RQ / ARQ yoki boshqa queue.

File Storage

S3 Compatible Storage.

Misol:

- MinIO;
- Cloudflare R2;
- AWS S3.

Infrastructure

Docker

Docker Compose

Nginx

Linux VPS / Cloud

Monitoring

Application logs

Error tracking

Health monitoring

5. TIZIM ARXITEKTURASI

Frontend

↓

REST API

↓

FastAPI Backend

↓

PostgreSQL

↓

Redis

↓

Background Workers

↓

Telegram / SMS / boshqa integrations

6. ASOSIY MENU

Chap sidebar juda kichik bo'ladi.

Bosh sahifa

Sotuv

Buyurtmalar

Mijozlar

Ombor

Kassa

Supplierlar

Hisobot

Sozlamalar

Oddiy sotuvchi uchun keraksiz bo'limlar yashiriladi.

7. HEADER

Har bir sahifada yuqorida:

Global Search

Qidirish mumkin:

- telefon;
- ism;
- buyurtma №;
- chek №;
- SKU;
- barcode;
- mahsulot modeli.

Shortcut:

Ctrl + K

Global qidiruv ochiladi.

8. BOSH SAHIFA — XODIM

Oddiy xodim kirganda juda sodda ekran.

Katta tugmalar:

+ Sotuv

+ Buyurtma

+ Mijoz

Buyurtmalar

Pastda:

Bugungi buyurtmalar.

Tayyor buyurtmalar.

Olib ketilmagan buyurtmalar.

9. BOSH SAHIFA — DIREKTOR

Direktor dashboardi boshqa.

Ko'rsatkichlar:

Bugungi tushum

Bugungi foyda

Sotuv soni

Yangi mijozlar

Buyurtmalar

O'rtacha chek

Qarzdorlik

Ombor qiymati

Kam qolgan mahsulotlar

Kechikkan buyurtmalar

10. QUICK ACTION

Tizimning deyarli hamma joyida:



Quick Action tugmasi.

Bosilganda:

- Sotuv
- Buyurtma
- Mijoz
- Kirim
- Xarajat

chiqadi.

11. MIJOZ QIDIRISH

Yangi sotuv yoki buyurtmada birinchi ekran:

Telefon yoki ism

Misol:

901234567

Tizim avtomatik qidiradi.

Agar mijoz topilsa:

Aziz Akbarov

+998 90 123 45 67

Oxirgi xarid: 12.07.2026

Qarz: 0

so'ng:

Tanlash

Agar mavjud bo'lmasa:

+ Yangi mijoz

12. YANGI MIJOZ

Majburiy maydonlar juda kam.

Ism

Telefon

Qolganlari optional:

- familiya;
- tug'ilgan kun;
- jinsi;
- izoh.

Mijoz yaratish 10-15 soniyadan oshmasligi kerak.

13. CUSTOMER PROFILE

Mijoz kartasi:

Header

Aziz Akbarov

+998 90 123 45 67

VIP

Jami xarid:

8 250 000 so'm

Qarz:

500 000 so'm

Tabs:

Umumiy

Retseptlar

Buyurtmalar

Xaridlar

To'lovlar

Servis

Fayllar

14. CUSTOMER TIMELINE

Customer profilida timeline mavjud.

Misol:

10.08.2026

Buyurtma №1824 yaratildi.

10.08.2026

500 000 so'm to'lov qilindi.

12.08.2026

Buyurtma tayyor bo'ldi.

13.08.2026

Buyurtma mijozga topshirildi.

Bu mijoz bilan bo'lgan barcha aloqani tushunishni osonlashtiradi.

15. PRESCRIPTION

Prescription alohida entity.

Har bir retsept version sifatida saqlanadi.

Eski retsept edit qilinsa ham eski qiymat yo'qolmasligi kerak.

16. PRESCRIPTION FORM

OD — Right

SPH

CYL

AXIS

ADD

PD

Prism

Base

OS — Left

SPH

CYL

AXIS

ADD

PD

Prism

Base

Qo'shimcha:

Near PD

Distance PD

Prescription Date

Doctor / Optometrist

Note

17. RETSEPT UX

Eng ko'p ishlatiladigan:

SPH

CYL

AXIS

PD

default holatda ko'rinadi.

Kam ishlatiladigan:

ADD

PRISM

BASE

Qo'shimcha parametrlar

ichida bo'ladi.

Shu bilan forma qo'rqinchli ko'rinmaydi.

18. RETSEPT HISTORY

Mijozda:

10.08.2026

OD -2.75

OS -1.00

04.02.2025

OD -2.50

OS -0.75

ko'rinishida tarix bo'ladi.

Istalgan ikki retseptni solishtirish mumkin.

19. YANGI SOTUV — ASOSIY FLOW

STEP 1

Mijoz.

Optional bo'lishi mumkin.

STEP 2

Mahsulotlar.

Barcode scanner yoki qidiruv.

STEP 3

To'lov.

STEP 4

Sotuv yakunlanadi.

Maksimal sodda.

20. POS SCREEN

Chap tomon:

Mahsulot qidirish.

O'ng tomon:

Savatcha.

Misol:

Ray-Ban RB3447

1 × 850 000

Blue Cut Lens

2 × 300 000

Case

1 × 50 000

Subtotal:

1 500 000

Discount:

100 000

Total:

1 400 000

21. PAYMENT MODAL

TO'LOV

bosiladi.

Modal:

Naqd

Karta

Click

Payme

Bank

Qarz

Boshqa

Split payment qo'llab-quvvatlanadi.

Misol:

Naqd:

400 000

Karta:

600 000

Qarz:

400 000

22. SALE FINALIZATION

Sotuv tasdiqlanganda transaction ichida:

1. Sale yaratiladi.
2. SaleItems yaratiladi.
3. Payment yoziladi.
4. Stock ayriladi.
5. Cashbox yangilanadi.
6. Customer history yangilanadi.
7. Audit log yaratiladi.

Bittasi ishlamasa transaction rollback qilinadi.

23. ORDER FLOW

Ko'zoynak tayyorlash:

Mijoz

↓

Retsept

↓

Ramka

↓

Linza

↓

Qo'shimcha xizmat

↓

Narx

↓

Avans



Buyurtma

24. ORDER CREATION UX

Hammasini bitta uzun formada ko'rsatish kerak emas.

Stepper:

1. Mijoz
2. Retsept
3. Ramka
4. Linza
5. To'lov

Pastda doim:

Orqaga

Davom etish

25. ORDER SUMMARY

Yakuniy ekranda:

Mijoz:

Aziz

Ramka:

RB3447

Linza:

Blue Cut 1.60

Retsept:

OD -2.75

OS -1.00

Narx:

1 650 000

Avans:

650 000

Qoldiq:

1 000 000

Taxminiy tayyor:

12.08.2026

BUYURTMA YARATISH

26. ORDER NUMBER

Human readable.

Misol:

OPT-260810-0158

yoki sodda:

№15824

Database UUID alohida bo'ladi.

27. ORDER STATE MACHINE

Buyurtmalar faqat belgilangan yo'nalishda harakat qiladi.

DRAFT

↓

CONFIRMED

↓

IN_PROGRESS

↓

READY

↓

NOTIFIED

↓

DELIVERED

Alohida:

CANCELLED

28. ORDER STATUS BUSINESS RULE

DELIVERED buyurtma:

to'liq yopilgan hisoblanadi.

Agar qarzi bo'lsa tizim warning ko'rsatadi:

Mijozda 500 000 so'm qoldiq mavjud.

Permission bo'lmasa qarz bilan topshirib bo'lmaydi.

29. KECHIKKAN BUYURTMA

Agar:

`expected_ready_at < current_time`

va status:

READY yoki DELIVERED bo'lmasa

buyurtma:

KECHIKKAN

deb belgilanadi.

Dashboardda ko'rinadi.

30. PRODUCT TYPES

Products umumiy jadvali mavjud.

Type:

FRAME

LENS

CONTACT_LENS

ACCESSORY

SERVICE

OTHER

31. FRAME

Frames uchun qo'shimcha atributlar:

Brand

Model

Color

Material

Gender

Shape

Eye Size

Bridge

Temple Length

Barcode

Image

32. LENS

Lens attributes:

Brand

Model

Lens Type

Index

Material

Coating

SPH

CYL

Diameter

Color

Photochromic

Blue Cut

Polarized

33. CONTACT LENS

Attributes:

Brand

Model

SPH

CYL

AXIS

BC

DIA

Expiry Date

Pack Quantity

34. PRODUCT VARIANTS

Bitta mahsulotning turli parametrlarini alohida variant sifatida saqlash kerak.

Misol:

Blue Cut 1.56

SPH:

-1.00

-1.25

-1.50

-1.75

...

Har variantning qoldig'i alohida.

35. STOCK

Mahsulot qoldig'i to'g'ridan-to'g'ri product table ichida saqlanishi tavsiya qilinmaydi.

Alohida:

InventoryBalance

va

StockMovement

bo'ladi.

36. STOCK MOVEMENT TYPES

PURCHASE

SALE

RETURN_IN

RETURN_OUT

TRANSFER_IN

TRANSFER_OUT

DAMAGE

INVENTORY_CORRECTION

MANUAL_ADJUSTMENT

ORDER_RESERVE

ORDER_RELEASE

37. STOCK LEDGER

Har bir harakat:

product_variant_id

warehouse_id

quantity

movement_type

reference_type

reference_id

user_id

created_at

bilan saqlanadi.

38. STOCK RESERVATION

Buyurtma yaratilganda ramka darhol sotilgan deb hisoblanmasligi mumkin.

Avval:

RESERVED

bo'ladi.

Masalan:

Physical:

10

Reserved:

3

Available:

7

Buyurtma bekor qilinsa:

reservation release qilinadi.

39. NEGATIVE STOCK

Default:

ombor minusga tushishi mumkin emas.

Permission bilan administrator override qilishi mumkin.

Override audit logga tushadi.

40. INVENTORY SCREEN

Jadval:

Mahsulot

Variant

Barcode

Kirim narxi

Sotuv narxi

Real qoldiq

Reserved

Available

Min qoldiq

Holat

Filter:

Brand

Category

Supplier

Warehouse

Low Stock

Out of Stock

41. STOCK CARD

Mahsulotni ochganda:

Current stock

Kirimlar

Sotuvlar

Transfer

Damage

Inventory correction

tarixi ko'rinadi.

42. PURCHASE

Supplierdan kirim.

Purchase:

Supplier

Invoice Number

Date

Items

Total

Paid

Debt

Warehouse

43. PURCHASE ITEM

Product

Variant

Quantity

Unit Cost

Discount

Total

44. WEIGHTED COST

Mahsulot tannarxini boshqarishda MVP uchun:

Weighted Average Cost

ishlatish mumkin.

Alternativ:

FIFO.

System setting orqali kelajakda o'zgartirish mumkin.

45. SUPPLIER PROFILE

Supplier:

Name

Phone

Telegram

Address

Contact Person

Balance

Notes

Tabs:

Purchases

Payments

Debt

Products

46. SUPPLIER PAYMENT

Supplierga pul berilganda:

SupplierPayment yaratiladi.

Payment method

Amount

Cashbox

Comment

Date

47. CUSTOMER DEBT

Customer balance avtomatik hisoblanadi.

Mijoz:

Total Orders:

5 000 000

Payments:

4 500 000

Debt:

500 000

Qarz qiymatini qo'lda o'zgartirib bo'lmaydi.

Faqat transactionlardan hisoblanadi.

48. CASHBOX

Har filialda bir yoki bir nechta kassa bo'lishi mumkin.

Cashbox:

Name

Branch

Currency

Current Shift

49. CASH SHIFT

Xodim ish boshlaganda:

OPEN SHIFT

Opening Balance:

1 000 000

Kun tugaganda:

CLOSE SHIFT

System Expected:

8 400 000

Actual:

8 350 000

Difference:

-50 000

50. CASH MOVEMENTS

SALE_PAYMENT

CUSTOMER_DEBT_PAYMENT

SUPPLIER_PAYMENT

EXPENSE

REFUND

CASH_IN

CASH_OUT

51. EXPENSE

Expense:

Category

Amount

Payment Method

Cashbox

Employee

Description

Attachment

Date

52. EXPENSE CATEGORY

Misol:

Rent

Salary

Transport

Utility

Marketing

Office

Repair

Other

53. RETURN

Oldingi sale topiladi.

Qaysi item qaytishi belgilanadi.

Reason kiritiladi.

Mahsulot:

SELLABLE

yoki

DAMAGED

tanlanadi.

SELLABLE:

stockga qaytadi.

DAMAGED:

damage movement yaratiladi.

54. SERVIS / REPAIR

RepairOrder:

Customer

Product Description

Problem

Photo

Service

Price

Prepayment

Expected Date

Status

Assigned Employee

55. REPAIR STATUS

RECEIVED

DIAGNOSIS

IN_PROGRESS

READY

DELIVERED

CANCELLED

56. EMPLOYEES

Employee:

First Name

Last Name

Phone

Position

Branch

Status

Hire Date

57. USERS

User:

Employee ID

Username

Password Hash

Role

Branch

Last Login

Is Active

58. ROLES

SUPER_ADMIN

OWNER

MANAGER

OPTOMETRIST

SELLER

CASHIER

WAREHOUSE_MANAGER

59. PERMISSION SYSTEM

Permission misollar:

customer.view

customer.create

customer.edit

prescription.create

sale.create

sale.discount

sale.cancel

payment.create

stock.view

stock.adjust

purchase.create

report.finance

employee.manage

60. DISCOUNT PERMISSIONS

Masalan:

SELLER:

max 10%

MANAGER:

max 25%

OWNER:

unlimited

Limitdan yuqori bo'lsa:

Manager PIN / approval.

61. AUDIT LOG

Audit log:

user_id

action

entity_type

entity_id

old_values

new_values

ip_address

device

created_at

62. AUDIT QILINADIGAN MUHIM OPERATSIYALAR

Price change

Discount

Sale cancellation

Stock adjustment

Product deletion

Purchase change

Payment deletion/cancellation

Customer debt operation

Role change

Login attempts

63. DELETE POLICY

Asosiy moliyaviy obyektlarda hard delete ishlatilmaydi.

Soft delete / cancellation.

Masalan:

Sale:

CANCELLED

Payment:

VOID

Purchase:

CANCELLED

64. DATABASE CORE TABLES

organizations

id

name

settings

created_at

branches

id

organization_id

name

address

phone

timezone

active

warehouses

id

branch_id

name

active

customers

id

organization_id

first_name

last_name

phone

phone_normalized

birth_date

customer_type

notes

created_at

updated_at

prescriptions

id

customer_id

branch_id

optometrist_id

od_sph

od_cyl

od_axis

od_add

od_pd

od_prism

od_base

os_sph

os_cyl

os_axis

os_add

os_pd

os_prism

os_base

distance_pd

near_pd

notes

prescription_date

created_at

products

id

organization_id

type

name

brand

model

sku

barcode

sale_price

minimum_stock

active

created_at

product_variants

id

product_id

attributes_json

barcode

sku

sale_price

active

inventory_balances

id

warehouse_id

product_variant_id

physical_qty

reserved_qty

updated_at

Unique:

warehouse + variant

stock_movements

id

warehouse_id

variant_id

type

quantity

unit_cost

reference_type

reference_id

created_by

created_at

suppliers

id

organization_id

name

phone

address

notes

created_at

purchases

id

supplier_id

warehouse_id

status

invoice_no

total

paid

balance

created_by

purchase_date

purchase_items

id

purchase_id

variant_id

quantity

unit_cost

total

sales

id

branch_id

customer_id

cash_shift_id

number

subtotal

discount

total

paid

balance

status

created_by

created_at

sale_items

id

sale_id

variant_id

description

quantity

unit_price

cost_price

discount

total

orders

id

branch_id

customer_id

prescription_id

number

status

subtotal

discount

total

paid

balance

expected_ready_at

ready_at

delivered_at

created_by

assigned_to

notes

created_at

order_items

id

order_id

item_type

product_variant_id

description

quantity

unit_price

cost_price

total

payments

id

branch_id

customer_id

reference_type
reference_id
payment_method
amount
cashbox_id
status
created_by
created_at

cashboxes

id
branch_id
name
active

cash_shifts

id
cashbox_id
employee_id
opening_amount
expected_amount
actual_amount
difference

opened_at

closed_at

status

expenses

id

branch_id

cashbox_id

category_id

amount

description

created_by

created_at

employees

id

organization_id

branch_id

first_name

last_name

phone

position

active

users

id

employee_id

username

password_hash

role_id

last_login

active

roles

id

organization_id

name

permissions

id

code

description

role_permissions

role_id

permission_id

audit_logs

id

user_id

action

entity_type

entity_id

old_data

new_data

ip

created_at

65. DATABASE INDEXES

Muhim indexlar:

customers.phone_normalized

products.barcode

product_variants.barcode

products.sku

orders.number

sales.number

orders.customer_id

sales.customer_id

stock_movements.variant_id

stock_movements.created_at

payments.customer_id

payments.created_at

66. PHONE NORMALIZATION

Telefonlar bazada bir xil formatda saqlanadi.

Masalan:

+998901234567

User:

90 123 45 67

yoki

901234567

yozsa ham topilishi kerak.

67. API STRUCTURE

Base:

/api/v1

68. AUTH API

POST

/auth/login

POST

/auth/logout

POST

`/auth/refresh`

GET

`/auth/me`

69. CUSTOMER API

GET

`/customers`

POST

`/customers`

GET

`/customers/{id}`

PATCH

`/customers/{id}`

GET

`/customers/{id}/timeline`

GET

`/customers/{id}/balance`

GET

`/customers/search?q=`

70. PRESCRIPTION API

GET

`/customers/{id}/prescriptions`

POST

`/customers/{id}/prescriptions`

GET

`/prescriptions/{id}`

PATCH

`/prescriptions/{id}`

71. PRODUCTS API

GET

`/products`

POST

`/products`

GET

`/products/{id}`

PATCH

`/products/{id}`

GET

`/products/barcode/{barcode}`

72. STOCK API

GET

`/inventory`

GET

`/inventory/{variant_id}`

GET

`/inventory/{variant_id}/movements`

POST

`/inventory/adjust`

POST

`/inventory/transfer`

POST

`/inventory/count`

73. PURCHASE API

GET

`/purchases`

POST

`/purchases`

GET

`/purchases/{id}`

POST

`/purchases/{id}/confirm`

POST

`/purchases/{id}/cancel`

74. SALE API

POST

`/sales`

GET

`/sales`

GET

`/sales/{id}`

POST

`/sales/{id}/cancel`

POST

`/sales/{id}/return`

75. ORDER API

POST

`/orders`

GET

`/orders`

GET

`/orders/{id}`

PATCH

`/orders/{id}`

POST

`/orders/{id}/confirm`

POST

`/orders/{id}/start`

POST

`/orders/{id}/ready`

POST

`/orders/{id}/notify`

POST

`/orders/{id}/deliver`

POST

`/orders/{id}/cancel`

76. PAYMENT API

POST

`/payments`

GET

`/payments`

POST

`/payments/{id}/void`

77. CASH API

POST

`/cash-shifts/open`

POST

`/cash-shifts/{id}/close`

GET

`/cash-shifts/current`

GET

`/cashboxes/{id}/movements`

78. REPORT API

GET

`/reports/dashboard`

GET

`/reports/sales`

GET

`/reports/profit`

GET

`/reports/products`

GET

`/reports/customers`

GET

`/reports/debts`

GET

`/reports/stock`

GET

`/reports/employees`

79. GLOBAL SEARCH API

GET

`/search?q=`

Response:

customers

products

orders

sales

80. FRONTEND ROUTES

`/`

Dashboard

`/pos`

Sotuv

`/orders`

Buyurtmalar

`/orders/new`

Yangi buyurtma

`/orders/[id]`

Buyurtma

`/customers`

Mijozlar

`/customers/[id]`

Mijoz

`/inventory`

Ombor

`/products`

Mahsulotlar

`/purchases`

Kirimlar

`/suppliers`

Supplierlar

`/cash`

Kassa

`/reports`

Hisobot

`/settings`

Sozlamalar

81. ORDER LIST UI

Default:

Bugungi va aktiv buyurtmalar.

Columns:

Number

Customer

Phone

Created

Expected

Amount

Balance

Status

Employee

Quick filters:

Bugun

Yangi

Ishda

Tayyor

Kechikkan

Topshirilgan

82. KANBAN VIEW

Buyurtmalar uchun optional Kanban.

Columns:

Yangi

Ishda

Tayyor

Xabar berildi

Topshirildi

Drag & Drop bilan status o'zgartirish mumkin.

Lekin backend status validation qiladi.

83. AUTOCOMPLETE

Linza yoki ramka tanlanayotganda:

2-3 harf yozilishi bilan natija chiqishi kerak.

Qidiruv:

brand

model

barcode

SKU

parameters

orqali ishlaydi.

84. BARCODE SCANNER

Oddiy USB barcode scanner keyboard kabi ishlaydi.

Maxsus hardware SDK shart emas.

Barcode kelganda product qidiriladi va savatchaga qo'shiladi.

85. PRINTER

Tizim:

A4 invoice

Thermal receipt

Barcode label

print qilishga tayyor qilinadi.

86. RECEIPT

Chekda:

Optika nomi

Filial

Sotuv raqami

Sana

Kassir

Mahsulotlar

Summa

Chegirma

To'lov

Qoldiq

Mijoz

QR

87. TELEGRAM BOT — PHASE 2

Bot asosiy tizim emas.

Rahbar uchun:

/today

/sales

/stock

/orders

/debt

88. AUTOMATIC TELEGRAM REPORT

Har kuni kechqurun:

Bugungi tushum

Foyda

Sotuvlar

Buyurtmalar

Yangi mijozlar

Qarz

Kam qoldiq

rahbarga yuboriladi.

89. CUSTOMER NOTIFICATION

Notification channel:

SMS

Telegram

WhatsApp — integratsiya mavjud bo'lsa.

Status:

PENDING

SENT

DELIVERED

FAILED

90. NOTIFICATION TEMPLATE

ORDER_READY:

Assalomu alaykum, {customer_name}.

Sizning #{order_number} buyurtmangiz tayyor.

Qoldiq to'lov:

{balance} so'm.

91. PWA

Tizim desktopga app sifatida o'rnatilishi mumkin.

Icon bosiladi.

Browser toolbar ko'rinmaydi.

Xodim uchun haqiqiy dasturdek ishlaydi.

92. INTERNET UZILISHI

MVPda to'liq offline POS qilish shart emas.

Lekin:

form draft

localStorage / IndexedDB orqali vaqtinchalik saqlanishi kerak.

Internet qaytganda user ma'lumotni yo'qotmasligi kerak.

93. AUTOSAVE

Murakkab order form:

har bosqichda draft saqlaydi.

Browser yopilib qolsa:

Tugallanmagan buyurtmani davom ettirish

chiqadi.

94. PERFORMANCE

Customer search:

< 500 ms maqsad.

Barcode lookup:

deyarli instant.

Order save:

normal internetda bir necha soniyadan kam.

Dashboard:

aggregation / cache ishlatiladi.

95. PAGINATION

100 000 ta mijoz bo'lsa ham barcha ma'lumot bir martada yuklanmaydi.

Server-side:

pagination

filtering

sorting

ishlatiladi.

96. SECURITY

Password hash:

Argon2 / bcrypt.

JWT yoki secure session.

Refresh token.

HTTPS.

Rate limiting.

Role permission.

Audit.

Backup.

97. BACKUP

PostgreSQL:

Daily automatic backup.

Retention:

7 daily

4 weekly

3 monthly

kabi policy qo'llash mumkin.

Production DB va backup bir serverda qolib ketmasligi kerak.

98. BRANCH SYSTEM

Barcha muhim entity:

organization_id

va kerakli joyda:

branch_id

bilan ishlaydi.

Bu sistema keyinchalik bir nechta filialga o'tishini osonlashtiradi.

99. BRANCH RESTRICTION

Oddiy sotuvchi:

faqat o'z filialini ko'radi.

Direktor:

barcha filiallarni ko'radi.

100. MULTI-WAREHOUSE

Bir filialda:

Main Warehouse

Shop Shelf

Lab

kabi bir nechta warehouse qo'llab-quvvatlanishi mumkin.

MVPda har filial uchun bitta default warehouse yetarli.

101. SETTINGS

Organization

Branches

Warehouses

Cashboxes

Payment Methods

Order Status

Expense Categories

Discount Limits

Receipt

Notifications

Users

Roles

102. REPORT — SALES

Filter:

Date

Branch

Seller

Product Type

Payment Method

Output:

Revenue

Number of Sales

Average Check

Discount

Cost

Gross Profit

103. GROSS PROFIT

Gross Profit:

Sales Revenue

-

Cost of Goods Sold.

Xarajatlar alohida.

Net Profitni to'liq buxgalteriya sifatida ko'rsatish MVP vazifasi emas.

104. PRODUCT ANALYTICS

Top selling.

Low selling.

No sale 30 days.

No sale 60 days.

No sale 90 days.

Stock turnover.

Low stock.

Out of stock.

105. CUSTOMER ANALYTICS

New Customers

Repeat Customers

Average Customer Spend

Customer Lifetime Purchases

Inactive Customers

Debt Customers

106. EMPLOYEE ANALYTICS

Seller:

Revenue

Sales count

Average check

Discount amount

Returns

Orders

107. NOTIFICATION CENTER

Headerda bell icon.

Masalan:

5 ta tayyor buyurtma.

3 ta kechikkan buyurtma.

8 ta kam qolgan mahsulot.

2 ta customer qarzi.

108. ERROR UX

Xodimga:

HTTP 500

ko'rsatish mumkin emas.

Masalan:

✗ Sotuv saqlanmadi.

Internet aloqasini tekshirib qayta urinib ko'ring.

yoki:

✗ Omborda bu mahsulotdan yetarli qoldiq yo'q.

Mavjud: 1 dona.

109. CONFIRMATION UX

Har narsada confirmation berish kerak emas.

Faqat xavfli operatsiyalarda:

Sotuvni bekor qilish.

Stock adjustment.

Refund.

Purchase cancellation.

110. COLOR UX

Statuslar butun sistemada bir xil.

Yangi — neutral.

Jarayonda — sariq.

Tayyor — yashil.

Kechikkan — qizil.

Topshirilgan — ko'k/neutral.

Rangga qo'shimcha matn ham doim bo'ladi.

111. RESPONSIVE

Tablet POS:

juda yaxshi ishlashi kerak.

Telefon:

rahbar monitoring uchun.

Desktop:

barcha funksiyalar.

112. MVP V1

Majburiy:

Authentication

Roles

Branches

Customers
Prescriptions
Products
Frames
Lens variants
Inventory
Stock movements
Suppliers
Purchases
Sales/POS
Orders
Payments
Customer Debt
Supplier Debt
Cashbox
Cash Shift
Expenses
Basic Reports
Audit Log
Backup

113. V1.1

Keyingi:

Returns

Inventory count

Barcode label

Repair

Advanced reports

Excel import/export

114. V2

Telegram bot

SMS

Customer reminders

Loyalty

Bonus

Appointments

Laboratory

QR tracking

Advanced analytics

Multi branch transfers

115. MVPGA KIRITMASLIK KERAK BO'LGAN NARSALAR

Boshlanishida keraksiz murakkablik bermaslik uchun:

AI recommendation

murakkab accounting

HR payroll

full offline architecture

mobile native app

machine learning

murakkab loyalty

marketing automation

majburiy emas.

Avval real optika ishini mukammal ishlatish kerak.

116. DEVELOPER MODULE STRUCTURE

Backend:

app/

core/

auth/

users/

customers/

prescriptions/

catalog/

inventory/

suppliers/

purchases/

sales/

orders/

payments/

cash/

expenses/

reports/

notifications/

audit/

integrations/

117. API RESPONSE STANDARD

Successful:

```
{ "success": true, "data": {} }
```

Validation error:

```
{ "success": false, "error": { "code": "VALIDATION_ERROR", "message": "...", "fields": {} } }
```

118. IDEMPOTENCY

Muhim payment va sale endpointlarda double click natijasida ikki marta transaction yaratilmasligi kerak.

Frontend idempotency key yuboradi.

119. DATABASE TRANSACTION

Sale yaratish:

bitta DB transaction.

Order confirm:

bitta DB transaction.

Stock transfer:

bitta DB transaction.

Purchase confirm:

bitta DB transaction.

Bu moliyaviy va inventory xatolarini kamaytiradi.

120. ACCEPTANCE TEST — CUSTOMER

Xodim telefon raqam yozadi.

Existing customer <1 soniyada topiladi.

Yangi customer minimal:

ism + telefon

bilan yaratiladi.

121. ACCEPTANCE TEST — SALE

Barcode scan.

Mahsulot savatchaga tushadi.

To'lov kiritiladi.

Sale yaratiladi.

Stock kamayadi.

Cashbox oshadi.

Customer history yangilanadi.

Audit yaratiladi.

122. ACCEPTANCE TEST — ORDER

Customer tanlanadi.

Prescription tanlanadi.

Frame tanlanadi.

Lens tanlanadi.

Avans olinadi.

Order yaratiladi.

Stock reserve qilinadi.

Order status kuzatiladi.

READY bo'lganda notification yaratilib,

DELIVEREDda order yopiladi.

123. ACCEPTANCE TEST — PURCHASE

Supplier tanlanadi.

10 dona mahsulot kirim qilinadi.

Purchase confirm.

Stock:

+10.

Supplier debt kerak bo'lsa avtomatik yaratiladi.

124. ACCEPTANCE TEST — CANCEL

Sale cancel qilinadi.

Stock qaytariladi.

Cash correction yaratiladi.

Sale history qoladi.

Audit log qoladi.

Original record o'chmaydi.

125. ENG MUHIM UX KPI

Oddiy yangi sotuv:

3 asosiy bosqich.

Oddiy customer:

2 maydon.

Barcode sale:

klaviaturada matn yozish talab qilmasligi kerak.

Existing customer:

telefon orqali topiladi.

Order status:

1 tugma bilan o'zgaradi.

126. OPTIKA OSNING IDEAL KUNLIK ISHLASHI

08:55

Xodim keladi.

Sistema ochiladi.

Smenani ochish

bosiladi.

09:20

Mijoz keladi.

Telefon:

901234567

Tizim topadi.

Oldingi retsept avtomatik ko'rinadi.

09:25

Yangi retsept kiritiladi.

Ramka barcode qilinadi.

Linza tanlanadi.

09:28

Buyurtma:

1 500 000.

Avans:

500 000.

Qoldiq:

1 000 000.

BUYURTMA YARATISH

09:29

Order:

№1582

yaratiladi.

Ramka:

Reserved.

12:00

Usta:

Ishda

qiladi.

16:20

Tayyor

bosadi.

Mijozga xabar yuboriladi.

18:00

Mijoz keladi.

Qoldiq:

1 000 000

to'laydi.

Topshirildi

bosiladi.

21:00

Direktor telefonidan dashboardni ochadi.

Bugungi tushum:

18 450 000

Foyda:

6 200 000

Buyurtma:

32

Tayyor:

8

Kechikkan:

1

Kam qoldiq:

12

Hammasini bir ekranda ko'radi.

127. YAKUNIY MAQSAD

Optika OS o'rnatilgan optikada quyidagi savollar uchun daftar, Excel yoki xodimdan so'rash kerak bo'lmasligi kerak:

Bugun qancha pul ishladik?

Kassada qancha pul bor?

Omborda qancha mahsulot bor?

Qaysi linza tugayapti?

Qaysi ramka yaxshi sotilyapti?

Qaysi buyurtma tayyor?

Qaysi buyurtma kechikdi?

Mijoz nima olgan?

Mijozning retsepti qanday?

Mijoz qancha qarzdor?

Suplierga qancha qarzmiz?

Qaysi xodim qancha sotuv qildi?

Qaysi tovar 3 oydan beri sotilmayapti?

Hammasi tizimdan bir necha soniyada olinishi kerak.

128. ASOSIY PRODUCT PHILOSOPHY

Less typing.

Less clicking.

More automation.

Xodim uchun oddiy.

Rahbar uchun kuchli.

Developer uchun tartibli.

Biznes uchun nazorat qilinadigan.

Bu — Optika OS.